Presentation 1

Link to presentation PDF, and code:

Presentation:
https://nuwildcat-my.sharepoint.com/personal/akl0407_ads_northwestern_edu/_layouts/15/stream.aspx?id=%2Fpersonal%2Fakl0407_ads_northwestern_edu%2FDocuments%2FBack%20up%2F2026-27%2FSTAT390%2FStudent%20presentations%2FPresentation%201%2FRecording1_Team3%2Emp4&referrer=StreamWebApp%2EWeb&referrerScenario=AddressBarCopied%2Eview%2E1ffcdcb8-8755-4d56-b6ac-7bd60ecc172f

PDF:
https://github.com/arvindkrishna87/Team3_DivideNConquer/blob/main/presentation_1/STAT%20390%20-%20Team%203%20Presentation%201.pdf

Code:
https://github.com/arvindkrishna87/Team3_DivideNConquer/tree/main/presentation_1/Code1_ModelTraining_23Oct_JingyuHuo

https://github.com/arvindkrishna87/Team3_DivideNConquer/tree/main/presentation_1/Code1_ModelTraining_23Oct_YifanXia

What did each of the team members do?

Note that multiple team members can work together on the same task. In that case, you may write the same task for all of them.

Ronan (Mathlete): I went over the main code files in the Presentation 1 code folder to look at the mathematical structure and how things were working in relation to the Mathlete objectives. I then pasted some code into Claude to make some nice visualizations for the slides

Jingyu (Coder): I reviewed all the files under the Presentation 1 code folder and understood the purpose of each function and class. I executed all stages 3 and 4 across five splits and got the result for each split. For the presentation, because me and Yifan both chose coder track so we splitted the work. I focused on the critical reflection part and identified three issues inside the current code and proposed feasible suggestion for each of them.

Yifan (Coder): I reviewed the complete pipeline and created a flowchart linking each processing step to its corresponding scripts. Using the supplied precomputed features, I executed Stages 3 and 4 across five train/validation/test splits with seed 3902026. In my personal copy, I enabled seeded shuffling and verified class proportions, case separation, and related-case grouping. I trained a fresh model for each split and

summarized accuracy, recall for both classes, confusion matrices, predicted probabilities for all 93 test cases, training/validation loss curves, and execution times.

Taylor (Analyst): I executed the attention_analysis.py pipeline across all folds to extract hierarchical attention weights across patches, slices, and stains (focused on Fold 5). I then conducted diagnostic and visual error analysis on four representative cases of case 12, 2, 78, and 18 by generating slice, stain, and whole-case visualizations of top and bottom attended patches. Finally, I documented methodological pipeline limitations regarding file indexing mismatches and attention correlation vs. causation.

How does it help the project (not required for presentation 1)?

Ronan:

Jingyu: These suggestions give us a clear starting point for the next round of code changes. We can then test whether those changes improve correctness and efficiency.

Yifan: These experiments establish a documented baseline for comparing future model changes. The five splits show how performance varies with data partitioning, with mean test accuracy of 83.80%.

Taylor: It informs future modeling and preprocessing work, specifically highlighting potential fixes concerning indexing and future testing to verify causation.

Issues faced that haven't been resolved:

Ronan: Nothing explicit, but I will do more research about my proposed fix to the class weight cancellation in the loss function issue, there may be a better fix.

Jingyu: Ready for code modification

Yifan: Training loss approached zero across all five splits, while validation loss plateaued or increased, suggesting overfitting.

Taylor: I identified a potential index issue between data loading and visualization scripts. Specifically, data_utils.py sorts patch file paths lexicographically as strings, whereas attention_analysis.py sorts numerically by parsed integers. Additionally, unhandled drops or caps of empty slices during feature extraction can shift tensor positions relative to raw PNGs, risking incorrect image lookups.

Proposed next steps (not mandatory for presentation 1)**:**

Ronan:

Jingyu:

Yifan: Modify the evaluation step to reload the checkpoint selected using validation loss before testing. Repeat the five experiments using the same splits and training settings, and compare the results with the current baseline to assess the effect of checkpoint selection.

Taylor: Potentially editing dataset and analysis scripts to share a single file index mapping to get rid of string vs. numeric sorting discrepancies. Also, potentially running tests to determine if high attention causally drove the final classification or if it merely reflects visual correlation (through systematically removing or masking the embeddings of top-attended patches).

References

Whose work did you build up on?

Ronan: Lecture, Code Walkthrough, CNN in CMIL, Multiple Instance Learning, Presentation 1 base code

Jingyu: Lecture, Code Walkthrough, Multiple Instance Learning, and CMIL Classification Problem Statement and Data

Yifan: Lecture, Code Walkthrough, Multiple Instance Learning, CNN in CMIL, and CMIL Classification Problem Statement and Data

Taylor: Lecture, CMIL Classification Problem Statement & Data; Lecture, Code Walkthrough; Presentation 1 base code